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First Deepwater Well Intervention with Hydraulic Workover Unit on a Tension Leg Wellhead Platform in Brazil

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Abstract

Production from a tension leg wellhead platform (TLWP) in the deepwater environment of Brazil led the operator to seek cost-effective well workover solutions after several projects were deemed uneconomical. The novel approach to diagnose, design, and deliver a customized hydraulic workover unit (HWU) solution to successfully change out a complex electrical submersible pump (ESP) completion on a TLWP with deck capacity and personnel restrictions is discussed.

Using HWU for ESP changeouts from offshore production platforms would be considered a regular operation, but its application in a deepwater TLWP environment remains uncommon. According to Pollack et al. (2007), previous deepwater offshore operations on live wells have only been carried out in the Gulf of Mexico. To ensure technical feasibility for HWU installation and operation on a TLWP, the engineered customization of the HWU solution was required. This effort involved technical visits, structural studies, and a detailed analysis of the intervention scope to diagnose challenges and design a solution that met the economic objectives.

The delivery of a customized HWU solution, tailored for the deepwater TLWP environment, enabled the successful changeout of an ESP well, which restored well production to 2,000 BOPD and achieved a 75% cost reduction compared to a conventional rig. To achieve this result in a challenging environment, the delivery of a customized design and installation of a modular and lightweight pulling unit that incorporates a hydraulic workover jack, load transfer system, skidding system, and temporary decks to enable the removal of tubulars from the wellbore was required. Operational challenges included blowout preventer (BOP) stabilization during adverse weather conditions while attempting an overpull requirement for unseating tubing hangers that exceeded HWU capacity. This situation was mitigated through the pre-planned integration of specialized equipment designed to provide a pull boost, which was successfully used to resolve the issue.

The deployment of a HWU solution on a deepwater TLWP, to include problem diagnosis, solution design, and operational execution, is discussed in this case study. The objective is to establish a benchmark for similar operations and demonstrates the HWUs capability to deliver efficient and cost-effective interventions under difficult conditions with consistent operational efficiency.

Introduction

Chacon et al. (2017) pointed out artificial lift systems, particularly ESPs, as a critical strategy to achieve target reservoir production and sustain economic efficiency. However, like any complex downhole equipment, ESPs require periodic maintenance and replacement to ensure optimal long-term performance through surface retrieval. Conventional workover rigs have traditionally supported these interventions, but HWUs provide a more efficient and cost-effective alternative with reduced requirements for logistical resources.

Duthie et al. (2019) recognized HWU as an effective alternative, particularly in environments with limited deck space, restricted load capacity, and reduced personnel availability. Its modular design, reduced footprint, and lightweight construction provide apparent advantages in deepwater environments that involve TLWPs, which impose significant structural and operational limitations.

Previous operations using HWUs on live wells have been successful in the Gulf of Mexico; however, this approach remained untested in Brazil, particularly in deepwater environments. This gap prompted the development and execution of a custom HWU solution designed to address the unique constraints of a Brazilian TLWP setting.

Lessons learned during the replacement of an ESP using a tailored HWU system engineered to overcome structural, logistical, and operational challenges are presented. The successful execution restored well production and delivered a substantial cost reduction compared to conventional alternatives, which established a new benchmark for offshore ESP interventions under limited conditions.

Statement of Theory and Definitions

D'Souza et al. (2013) describe TLWPs as floating offshore platforms anchored by vertical tendons designed to support dry x-mas tree systems in deepwater environments (Fig. 1). However, their limited deck space and restricted personnel capacity create significant challenges for conventional rig-based interventions. In this context, an HWU provides a practical and cost-effective alternative.

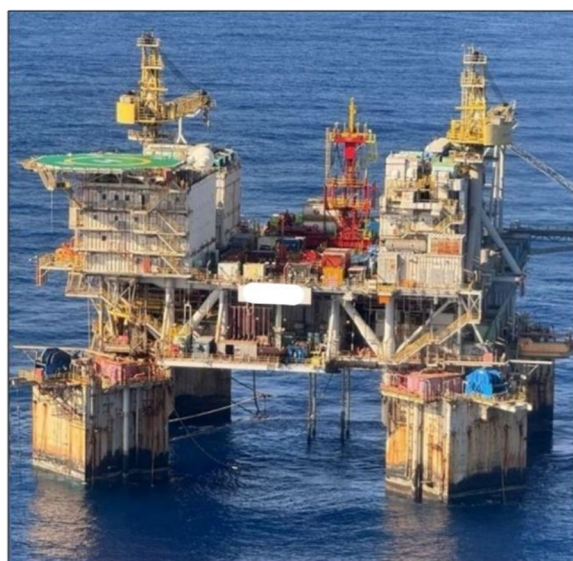


Figure 1—HWU at TLWP.

Maddox (1994) defines HWUs as a method that hoists or snubs (when pushing under pressure) oilfield tubulars into and out of wellbores. The technology operates in both onshore and offshore environments and uses hydraulic cylinders to manage tubulars. HWUs with a smaller footprint and modular design perform

well interventions similar to conventional rigs and help prevent the necessity for large support vessels or heavy infrastructure.

During standard operations, the HWU connects mechanically through the BOP stack to the wellhead, a common practice for both live and dead well operations. Offshore applications often require a substructure to transfer weight from the wellhead to the platform's capping beams. The use of this solution depends on the operator's requirements and crane availability on the offshore platform. A conventional HWU setup includes (Fig. 2):

- Gin pole with winches: A vertical mast equipped with winches that moves the bottomhole assembly (BHA) and long tubulars between the deck and well center.
- Work basket: The primary working platform in HWUs. It accommodates the jack operator, winch operator and power tong operator, along with all their control consoles. At the work basket level, key operations such as make-up and break-out tubular connections are performed.
- Jack with a pair of hydraulic slips: The primary lifting mechanism of the HWU. It operates in with a set of stationary and travelling (mounted on the jack's head) hydraulic slips. The alternate grip between the stationary and travelling ensures continuous and controlled pipe movement during intervention.
- Work window: Alternative structure installed to accommodate tubulars or BHAs with large outer diameters (OD) or insufficient length for safe jack operations.
- Substructure and skidding system: Substructure is designed to distribute the weight from the well head to the deck beams at the platform. It also enables HWU movement between well slots without rig down, making it ideal for multi-well operations.
- Anchoring system: Stabilization system essential for the HWU high vertical structure, especially considering windy scenarios. There are basically two types of anchoring system: guy wires attached to platform deck or outriggers within the structure itself.
- WHPCE: Assembly that includes the BOP set, valves, accumulator unit, choke and kill lines, manifold and control valves to manage well pressure during interventions and maintain safety and well integrity during interventions.
- Accumulator System (Koomey): An independent hydraulic control system that stores pressurized energy to operate the BOP and other critical WHPCE components in case of emergency or loss of main power. It is vital for ensuring well control and safety.
- Hydraulic power unit: Core component that supplies hydraulic energy to the jack, slips, and winches.
- Fluid system: Set of pumps, high-pressure lines, and tanks used to manage fluid operations during interventions.

Anyachie et al. (2024) note that HWUs, originally developed during the 1920s for land-based operations, provide a safer and more controlled method to snub and pull tubulars under pressure. The technology later evolved to address offshore demands, particularly in environments where conventional rigs prove impractical. Its modular design, small footprint, and easy mobilization make HWUs well suited for TLWPs, where structural and logistical constraints remain significant.

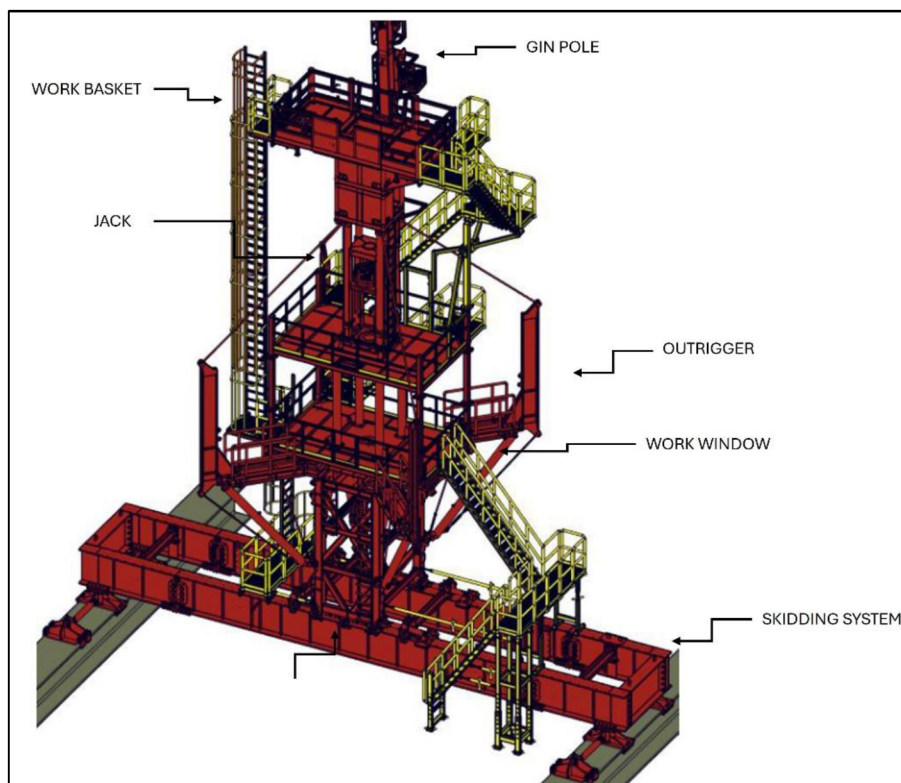


Figure 2—HWU components.

Engineering Assessment and Operational Setup

A tender-assisted drilling (TAD) rig previously performed well operations on the TLWP platform. This setup proved effective but incurred high operational costs, primarily because of the requirement for a separate tender vessel and the TAD rig's complex anchoring system. After a new operator acquired the field and focused on resource optimization, the HWU replaced the TAD rig to perform ESP changeouts using only the platform's existing resources.

To meet project-specific challenges, a tailored solution was developed that matched the unique requirements of the TLWP platform. The operator prioritized a solution that fit within the limited deck space and helped prevent the requirement for additional resources such as support vessels. The solution extended beyond HWU rig equipment and included auxiliary equipment such as ESP cable drums and 1,800 bbl of fluid storage. The decision to have this high-volume capacity onboard was driven by the logistical complexity of the offshore operations because it was important to ensure sufficient fluid storage to maintain operations for up to three days if the formation isolation valve failed to actuate and remain open during the ESP changeout.

To enable this, a detailed structural study of the TLWP deck analyzed available space and the deck's load capacity (Fig. 3). This study considered the weight and dimensions of all HWU components, required auxiliary equipment, and live loads expected during ESP changeouts. As a result, three auxiliary decks for active tanks, storage tanks, and the pipe rack area were designed and constructed (Fig. 4). Simulations and modeling software supported the design process to develop an optimized equipment layout (Fig. 5). This approach ensured that the solution fit within the available space and met all structural load requirements.

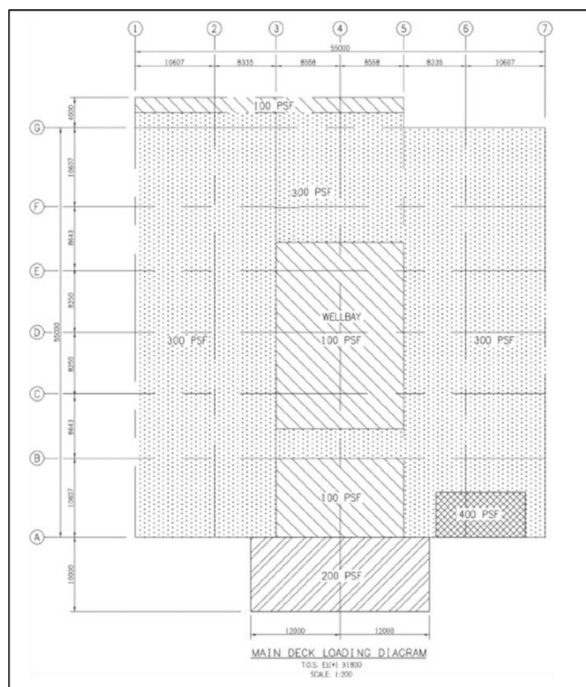


Figure 3—Main deck load capacity.

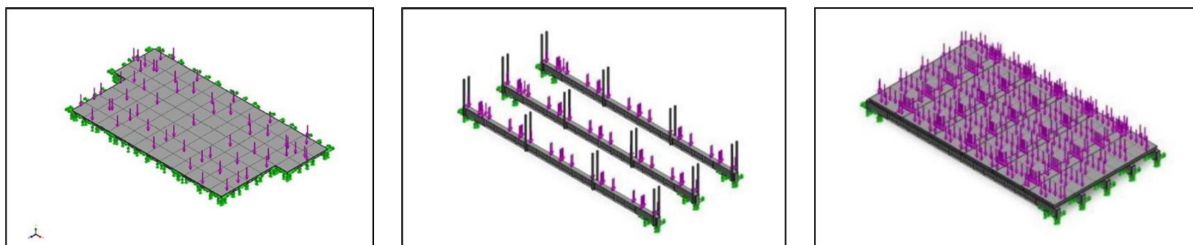


Figure 4—Auxiliary deck model.

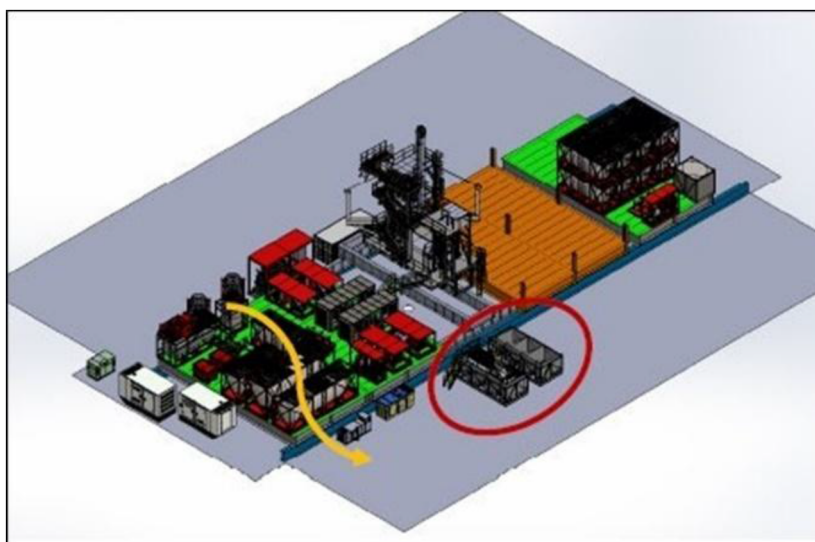


Figure 5—Equipment layout.

The original plan specified a 340,000-lbm capacity HWU to accommodate the platform's space and load limits. However, well handover data and trip-out simulations using torque and drag software predicted an

overpull of approximately $\pm 60,000$ lbm to unset the tubing hanger, potentially exceeding the unit's rated capacity (Fig. 6 and Fig. 7). The HWU structure was upgraded to support loads up to 400,000 lbm to meet this requirement.

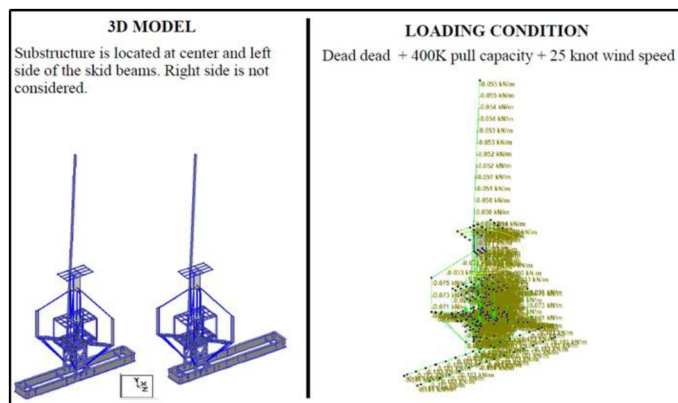


Figure 6—HWU loading model.

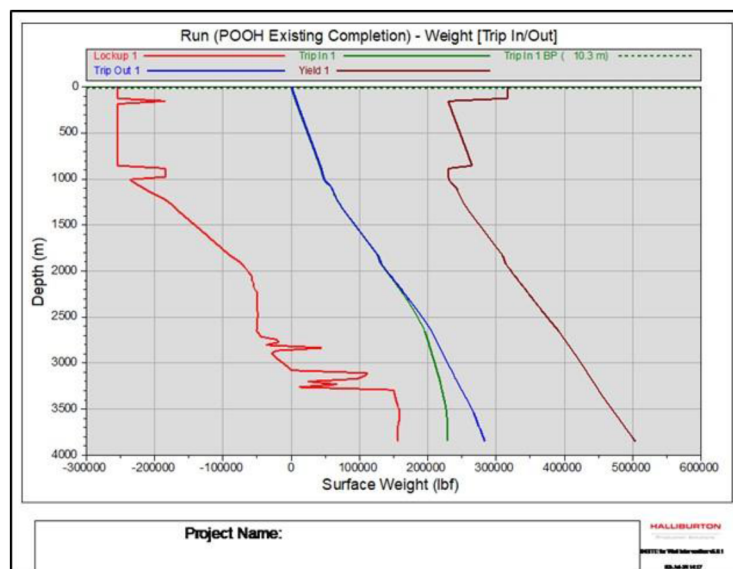


Figure 7—Software POOH simulation.

Meeting this requirement also necessitated the integration of a pull-boost system to provide the additional pulling force beyond the HWU's hydraulic jack capacity. To facilitate the pull boost, a current tension table system was modified to a C-plate configuration to allow greater flexibility within the limited HWU space work window. To ensure the modified structure could safely accommodate the expected loads, detailed stress simulations using computer software validated and optimized the design for both performance and safety (Fig. 8).

Furthermore, the operation required a 14-in. bore custom work window because of the complexity of the completion. The operation included a 13 3/8-in. tubing hanger, a 10.75-in. ESP completion shroud, and completion strings of various diameters. This configuration demanded a larger work window to support larger power tongs and to allow sufficient clearance and flexibility for safe and efficient management of all components. The structure also had to withstand the full 400,000-lbm pulling capacity to ensure compatibility with the upgraded unit.

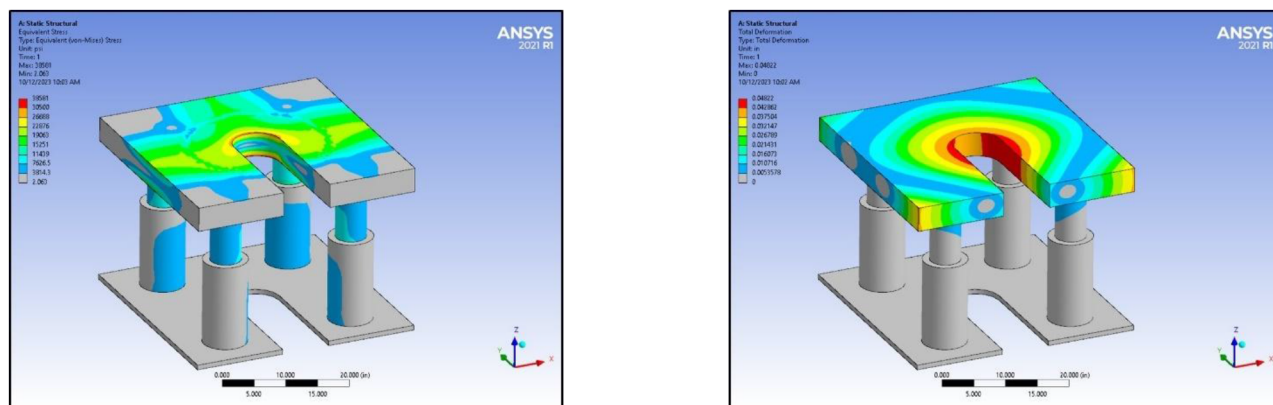


Figure 8—Software tension table simulation.

The available space did not allow the installation of the ESP motor lift extension (MLE) cable spooler, even with the work window in place. To address this, the team engineered, designed, and installed a support structure at the work window level to accommodate the MLE cable reel (Fig. 9). The necessity for this special structure arose during the HWU rig-up in Brazil because of specific cable requirements. This solution addressed the challenge effectively, which included issues identified at the last minute.

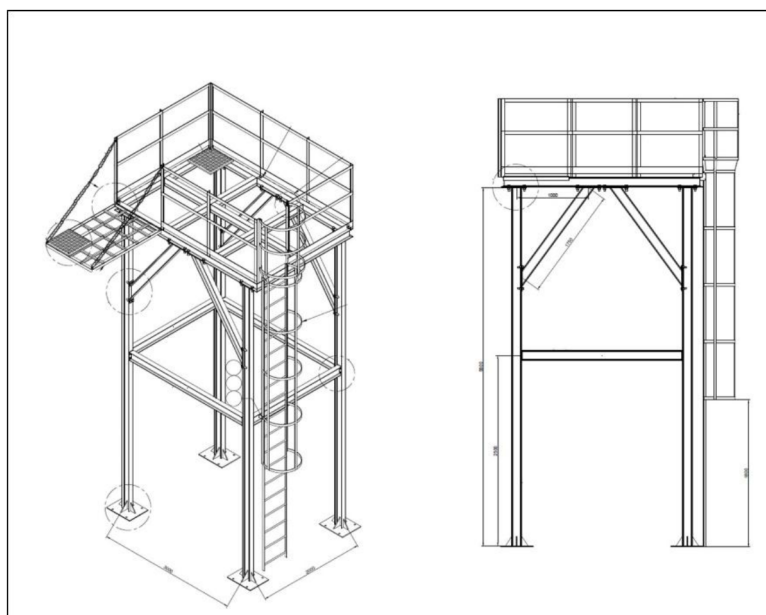


Figure 9—Support structure for the MLE cable reel.

These tailored engineering solutions ensured safe operations at the TLWP and maintained operational integrity throughout the intervention.

ESP Changeout

The ESP changeout campaign began after ESP failure, which created an urgent need to reestablish well production. Weight limitations at the wellhead required the installation of a custom substructure equipped with a skidding system to distribute unit and operational weight in current platform deck beams. The operation proceeded with the equipment rig up, and during this phase, an unexpected challenge emerged because of the specific characteristics of the TLWP. While the spacer risers and BOP stack on the wellhead were installed, significant lateral movement caused misalignment between the BOP and HWU bore.

This scenario was not anticipated as in previous operations, which used the TAD rig anchoring system and derrick to offset lateral movement and allow the TLWP to behave like a fixed platform. Without the TAD rig, this operation marked the first time where the riser's natural movement became noticeable during the BOP installation. To stabilize the system and allow operations to continue, even during bad weather, the riser tensioner rollers were tightened and adjusted, and a set of straps and shock absorber springs was added to the BOP stack (Fig. 10 and Fig. 11).

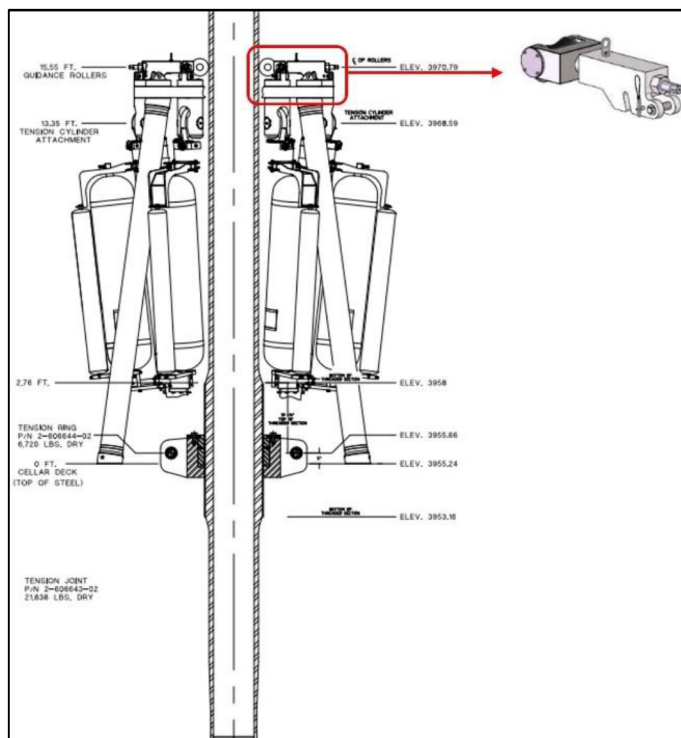


Figure 10—TLWP riser tensioners with rollers.

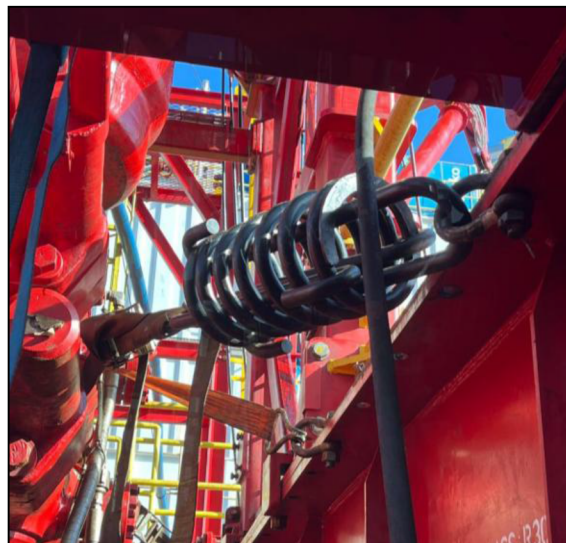
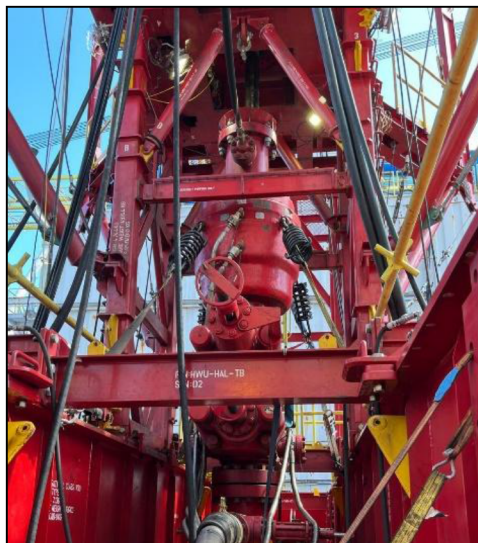


Figure 11—BOP stack with shock absorber spring and straps.

With the BOP stack stabilized, the team rigged up the rest of the equipment and performed function and pressure tests. Once all systems were confirmed as operational, well intervention commenced on the first well (Table 1). The major planned steps for the ESP changeout after BOP installation included:

1. Skid the unit over the BOP and pressure test the WHPCE.
2. Drain and clean the well annulus.
3. Retrieve the plug installed at the tubing hanger profile.
4. Clean the production string.
5. Retrieve the tubing hanger.
6. Pull out of hole with the old 10.75-in. ESP production string.
7. Run in hole with the new 7 5/8-in. ESP production string.
8. Install the tubing hanger and plug.
9. Rig down the BOP.
10. Skid the unit away from the wellhead.

Table 1—First well: old upper production string.

Length (m)	Description
1.6	13 3/8-in. tubing hanger
1900	7-in. production string
60	5.5-in. production string
15.9	4.5-in. production string
406.7	5.5-in. production string
42.5	10.75-in. ESP assembly
1.9	6 5/8-in. production string
76.7	5.5-in. production string
37	4.5-in. production string
398	5.5-in. production string
15	Shifter assembly

The pullout of the ± 2956 -m upper production string on the first well began with unseating the tubing hanger. This step required the tension table to apply an overpull beyond the HWU's 340,000-lbm pulling capacity. The tension table and manual slips were installed and 345,000 lbm was applied; the hanger was released and disassembled at the work window level because of the 13 3/8-in. OD (Fig. 12). After that, the hydraulic jack retrieved the completion string. The operation proceeded at a controlled pace to manage multiple diameters and components (i.e., ESP cables, the chemical injection mandrel, and the downhole safety valve) safely and without equipment damage.

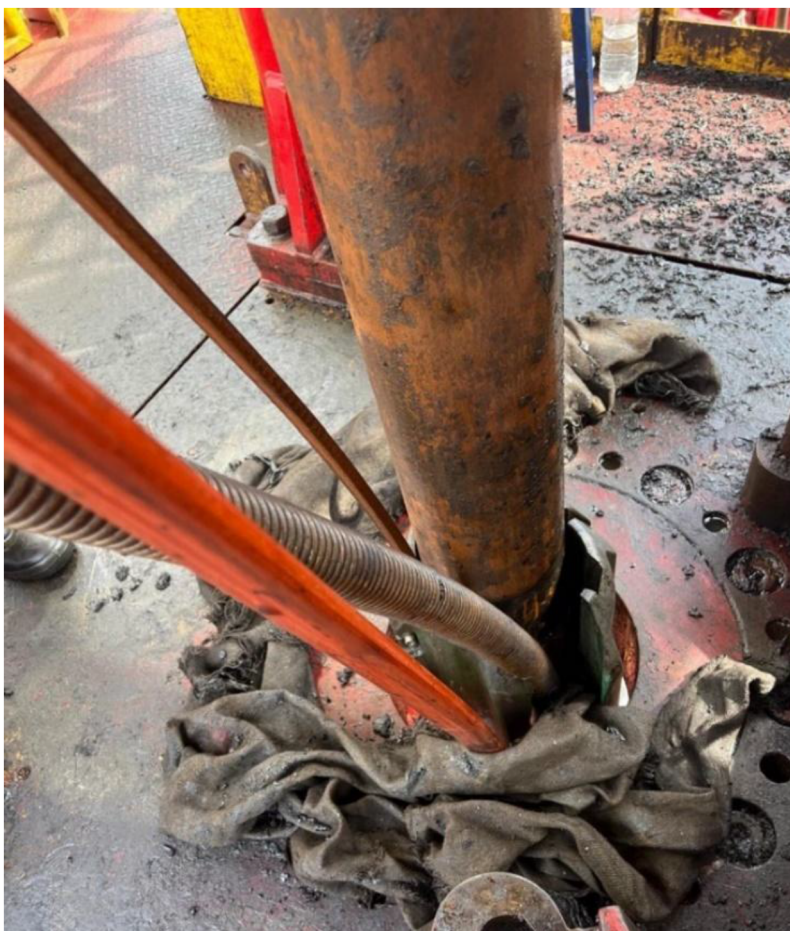


Figure 12—Cables at the work window level.

The 10.75-in. ESP operated outside the expected range, based on its performance curve. The operator decided to install a new production string with a 7 5/8-in. ESP to align with current well conditions and improve overall performance. The team installed the new string successfully and brought the well online, which restored stable oil production (Table 2).

Table 2—First well: new production string.

Length (m)	Description
1.6	13 3/8-in. tubing hanger
1412.5	5.5-in. production string
971.2	4.5-in. production string
47.5	5.5-in. production string
31.1	4.5-in. production string
62.73	7 5/8-in. ESP assembly
14.8	5.5-in. production string
527.9	4.5-in. production string
31.29	5.5-in. production string
30.6	Shifter assembly

Conclusions

The first successful deepwater well intervention with a HWU on a TLWP in Brazil is described in this paper. The team encountered and adapted to several new challenges throughout the operational phase. Both the HWU and the operational sequence required adaptation to the TLWP's unique characteristics, which demanded precise planning and operational flexibility.

One key lesson learned during the planning phase involved the necessity for an updated structural analysis of the deck before mobilization. This evaluation identified potential required structural changes, such as auxiliary deck installations, that would safely accommodate all mandatory equipment.

Another key consideration involves riser movement. HWUs operate on dry x-mas trees and lack a compensation system; therefore, riser movement should be accounted for. On a TLWP, the tensioner rollers should be properly adjusted and maintained to keep the HWU bore aligned with the wellhead. In some cases, particularly during bad weather, even with properly adjusted rollers, the BOP stack might require stabilization with shock absorbers or straps. During the design and fabrication phase of the substructure and skidding system, it is important to consider the possibility of skidding the HWU with live load. In specific situations, the unit should move during the intervention to maintain alignment with the well, and the substructure should be designed to safely support this condition.

A thorough review of all equipment to be installed or retrieved from the well remains essential. Certain component diameters and geometries might require structural adjustments, which includes pad eyes installation for sheaves, clearance for large hydraulic tongs, or accommodations for tension table use.

Despite the challenges, the diagnosis and implementation of solutions allowed for a successful completion of the ESP changeout, which restored well production to 2,422 B/D at 15% of the daily cost of a TAD rig (Fig. 13). The HWU solution fundamentally changed the operator's approach to well interventions through the replacement of traditional, high-cost methods with more efficient, customized service. This shift not only reduced operational costs but also allowed the client to perform interventions with greater flexibility and efficiency, challenging the industry's conventional practices and positioning our company as a strong player in the HWO market, especially in TLWP environments.

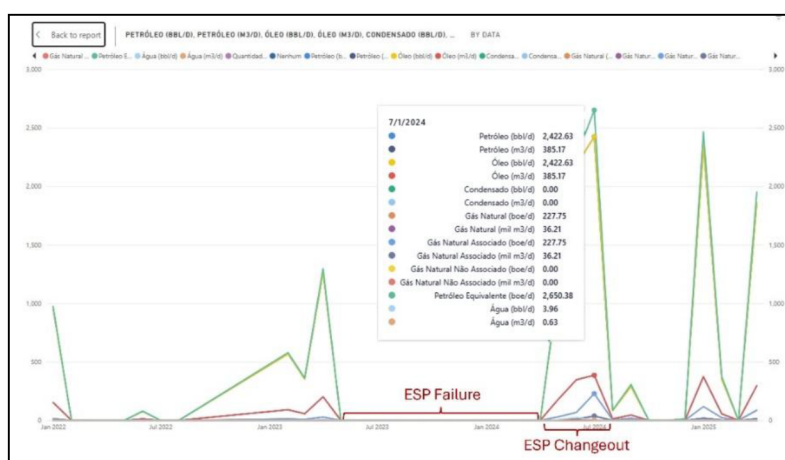


Figure 13—ANP data of well production.

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